

KEENER'S METHOD

AP

SUMMARY OF MASSEY'S METHOD

MASSEY'S VISION

Ratings are a unit quantity distributed among tournament participants.

The data that drives ratings is point difference.

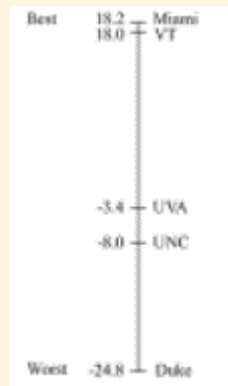
The difference in strength is latent but revealed by the points difference in a direct match.

By definition, points difference sums to 0; the natural linear algebra formalisation has a singular matrix and is not actionable.

Massey *alters* the matrix to guarantee that a solution exists, if approximate.

Massey's ratings are the solution $\mathbf{r}$ of $\overline{M}\mathbf{r} = \mathbf{p}$

Team	Rating r	Rank
Duke	-24.8	5th
Miami	18.2	1st
UNC	-8.0	4th
UVA	-3.4	3rd
VT	18.0	2nd



KEENER'S METHOD

KEENER'S VIEW, 1

One's *strength* should be measured relatively to their opponents'

Team i might be strong against team j but weak against k and so on:

$$s_i = \sum_{j=1}^m s_{ij}$$

where $s_{ii} = 0$ (i cannot play itself)

KEENER'S VIEW, 2

As with Massey, ratings are a unit quantity distributed among tournament participants:

$$\sum_{i=1}^m r_i = 1$$

Pie chart effect: one's rating improvement can only come as others' worsens.

Later, ratings will determine rankings and winning probabilities.

KEENER'S VIEW, 3

K. believes that strength, which is *manifest*, and rating, which is *latent*, should be connected by a scaling factor λ , which is to be determined for each league/tournament:

$$s_i = \lambda r_i$$

So, in vector notation:

$$\mathbf{s} = \lambda \mathbf{r}$$

At the moment we know neither of the three... let's start with strength.

THE INPUT DATA

K. does not commit to a specific way to gauge strength:

a_{ij} = the statistics produced by team i when playing j

non-negativity requirement: $a_{ij} \geq 0$

EXAMPLE STATS: WINS

Consider wins/ties:

$$a_{ij} = W_{ij} + \frac{T_{ij}}{2}$$

EXAMPLE STATS: POINTS

Points scored against:

$$a_{ij} = S_{ij}$$

points are considered a *crude* measure of strength.

avoid high-scoring matches to have a disproportionate effect by means of relative scoring:

$$a_{ij} = \frac{S_{ij}}{S_{ij} + S_{ji}}$$

THE LAPLACE CORRECTION

a *cold start* problem that is often encountered in Data Science:

at the start, lack of data makes the rating not meaningful or even impossible.

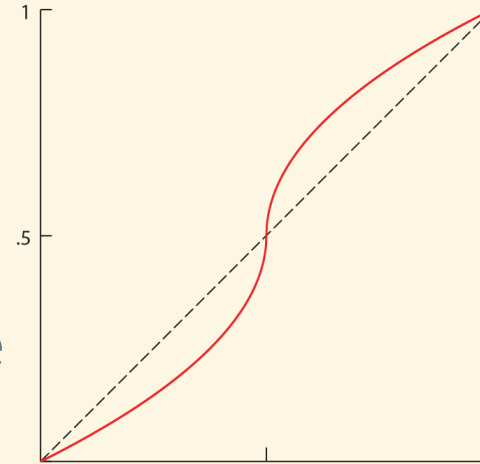
Laplace sets stats to 0.5, with minimal alteration of subsequent measures

$$a_{ij} = \frac{S_{ij} + 1}{S_{ij} + S_{ji} + 2}$$

if $S_{ij} \approx S_{ji}$ and both are large then $a_{ij} \approx \frac{1}{2}$ (Good or bad?)

SKEWING

- it mitigates convergence to $\frac{1}{2}$ over time
- it sterilises the effect of extreme scores



$$h(x) = \frac{1}{2} + \text{sgn}\{x - (1/2)\} \sqrt{|2x - 1|}/2$$

additionally, $a_{ij} \leftarrow \frac{a_{ij}}{n_i}$ to balance no. of games.

KEENER'S STRENGTH

Strength revealed by performance (scoring) but tempered by the strength of the opponent themselves.

Relative s. of i when playing against j :

$$s_{ij} = a_{ij} \cdot r_j$$

(N.B. *scoring* is S_{ij} while *strength* is s_{ij})

CUMULATIVE STRENGTH

Cumulative/absolute strength of team i :

$$s_i = \sum_{j=1}^m s_{ij}$$

$$\mathbf{s} = \begin{pmatrix} \sum_{j=1}^m s_{1j} \\ \sum_{j=1}^m s_{2j} \\ \vdots \\ \sum_{j=1}^m s_{mj} \end{pmatrix}$$

$$\mathbf{s} = \begin{pmatrix} \sum_{j=1}^m s_{1j} \\ \sum_{j=1}^m s_{2j} \\ \vdots \\ \sum_{j=1}^m s_{mj} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{pmatrix}$$

$$\mathbf{s} = \begin{pmatrix} \sum_{j=1}^m s_{1j} \\ \sum_{j=1}^m s_{2j} \\ \vdots \\ \sum_{j=1}^m s_{mj} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{pmatrix} = A\mathbf{r}$$

The *strength vector* $\mathbf{s}$ that collects all cumulative strengths is

$$\mathbf{s} = A\mathbf{r}$$

where $\mathbf{r}^T = \{r_1, \dots, r_m\}$ is the rating vector.

The argument has a certain circularity...

FINALLY

Since rating should be proportional to strength:

$$\mathbf{s} = \lambda \mathbf{r}$$

$$A\mathbf{r} = \lambda \mathbf{r}$$

So, rating really is an e-vector of A , and λ an e-value.

OBSERVATIONS

We would like a positive λ

also the values in $\mathbf{r}$ should be positive

In general, a *reasonable* solution is **not** guaranteed:

- which eigenvalue (among up to m) to choose?
- even for positive λ s the relative e-vector could contain negative or even complex numbers!

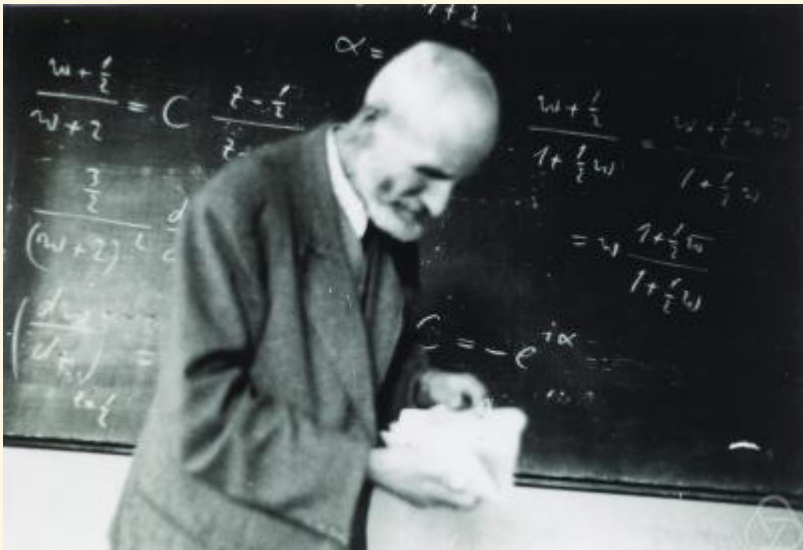
THE PERRON-FROBENIUS THEOREM

NON-NEGATIVITY

Perron-Frobenius focus on matrices that contain only non-negative values:

$$A = [a_{ij}] \geq 0$$

This is easily the case when a_{ij} is some *stats* on winning or scoring etc.



IRREDUCIBILITY

Perron-Frobenius request that each pair i, j be *connected*:

- simply, $a_{ij} > 0$ (i.e., teams have played before)
- or there is a non-negative path of p intermediate “steps” $k_1, \dots, k_p$:

$$a_{ik_1} > 0, a_{k_1k_2} > 0, \dots, a_{k_pk_j} > 0$$

IRREDUCIBILITY IN PRACTICE

it requiring that each teams has played common opponents in the past, even indirectly, e.g.:

$$a_{\text{Burnley,Nice}} = 0$$

but since

$$a_{\text{Burnley,Arsenal}} > 0, a_{\text{Arsenal,PSG}} > 0, a_{\text{PSG,Nice}} > 0$$

a tournament containing both Burnley and Nice is suitable.

Irred. may not hold at the beginning of a tournament but it's not considered **prohibitive**.

GOOD NEWS

If A is non-negative and irreducible, then

- the dominant e-value is real and strictly positive: our λ !
- except for positive multiples, there's only one non-negative e-vector $\mathbf{x}$ for A : (almost) our $\mathbf{r}$!
- the final $\mathbf{r}$ is obtained by normalizing $\mathbf{x}$: $\mathbf{r} = \mathbf{x} / \sum_j x_j$
- individual ratings r_i will be in $(0,1)$ and will sum to 1.

PERRON-FROBENIUS

Perron–Frobenius Theorem

If $\mathbf{A}_{m \times m} \geq \mathbf{0}$ is irreducible, then each of the following is true.

- Among all values of λ_i and associated vectors $\mathbf{x}_i \neq \mathbf{0}$ that satisfy $\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i$ there is a value λ and a vector $\mathbf{x}$ for which $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ such that

- ▷ λ is real.
- ▷ $\lambda > 0$.
- ▷ $\lambda \geq |\lambda_i|$ for all i .
- ▷ $\mathbf{x} > \mathbf{0}$.

- Except for positive multiples of $\mathbf{x}$, there are no other nonnegative eigenvectors $\mathbf{x}_i$ for $\mathbf{A}$, regardless of the eigenvalue λ_i .
- There is a unique vector $\mathbf{r}$ (namely $\mathbf{r} = \mathbf{x} / \sum_j x_j$) for which

$$\mathbf{A}\mathbf{r} = \lambda\mathbf{r}, \quad \mathbf{r} > \mathbf{0}, \quad \text{and} \quad \sum_{j=1}^m r_j = 1. \quad (4.11)$$

- The value λ and the vector $\mathbf{r}$ are respectively called the *Perron value* and the *Perron vector*. For us, the Perron value λ is the proportionality constant in (4.9), and the unique Perron vector $\mathbf{r}$ becomes our *ratings vector*.

OBSERVATIONS

- the conditions are strict but not impossible
- a strong memory effect makes Keener's ratings represent long-term tendencies
- today, random walks/Montecarlo methods approximate Keener's rating without the need to extract e-pairs of large matrices.
- [\[Keener, SIAM Review 35:1, March 1993\]](#) is credited with seeding the ideas behind Google's PageRank.